

I built a customer
risk model.
Then I tested it on
the customers who
actually caused UK
banks to be fined.

Saman Barati · London · September 2026

Excel and Power Query. No machine learning, on purpose.

Most risk models are judged on whether the maths works.

That is the easy test. A customer risk assessment can be perfectly consistent and still rate the wrong people Low.

So I built one properly — 20 risk factors mapped to regulation 18(2)(b) of the Money Laundering Regulations, 98 level definitions, every weight carrying a written rationale and a published source — and then rebuilt six customers named in FCA enforcement notices as they looked on the day each bank took them on.

Each one is scored at least twice, because a reconstruction from a published notice is twenty judgement calls and reporting one number hides nineteen of them.

Four are read once in the customer's favour and once at the worst reading the notice does not rule out. Two are pairs of a different kind: one scored on two dates eleven days apart, and one with and without a verification the FCA found was missing.

THE RESULT, AND THE PROBLEM WITH IT

Not one reaches High on the arithmetic.

In any reading. Three of the six are rated High, and every one of them through the same mandatory escalator rather than through a score.

3.40

is the highest score any of them *could* have reached, with every open factor pushed to 5. High begins at 3.50.

The facts each notice settles — a UK company with UK owners, or an ordinary UK personal customer — pin 40–69% of the model's weight at level 1 before any judgement is made. The headline finding was fixed by the inputs, not found by the test. What the back-test can still decide is whether an escalator fires.

FINDING ONE

One word decided the most serious case.

Fowler Oldfield was a Bradford jewellery business. Around £365m was deposited with NatWest, roughly £264m of it in cash. It produced the FCA's first criminal prosecution under the money laundering regulations.

Read as a cash-intensive trade	1.82 → Low
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Read as a dealer in high-value goods	1.90 → High
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Same file, same day. The scores differ by 0.075 and neither crosses a band. The rating differs completely, because one reading fires an escalator and the other does not.

That is a defect in my level definitions, not in the analyst reading them. So I wrote a fix.

My fix would have broken the only case the model got right.

Barclays opened an account for a gold refining and trading company in January 2015. The application form said £500,000 turnover, manually amended to £3 million, and that it would not trade outside the EU. Eleven days later the file recorded gold from Ghana and Burkina Faso.

My model rates it High on day one — 2.14, escalated — because "gold refining and trading" is a dealer in high-value goods under the same ambiguous definition that page four calls a defect.

Tie that level to *registration*, as I had recommended, and this customer drops to Medium. A gold trader declaring no cash would not have been registered.

I wrote a fix, tested it on the case that motivated it, and never asked what else it touched. The recommendation is still on the list, now with the argument against it attached.

FINDING THREE

A customer whose address was a London landmark scores 1.52.

Monzo onboarded customers who gave well-known London landmarks as their home address. My library has a factor written for exactly that — from that notice, before any of it was scored.

I then gave the factor **1.5% of the model**, and wrote a paragraph defending the weight.

Scored through, the customer lands comfortably in Low. The factor that saw the problem was too light to change the answer.

Both the reasoning and the result stay in the repository. Deleting the argument I got wrong would have removed the only evidence that the test worked.

FINDING FOUR

The weights were never the thing that mattered.

I spent a whole document arguing about whether geography should carry 25% or 15%.

Move any category weight by 10 percentage points	up to 26 of 400 customers change band
The same change, against band boundaries placed where the population actually sits	29 to 100 customers change band

The boundaries decide the output. The weights barely touch it.

Which reverses the order of the work: fix the bands first, argue about the weights second. Those boundaries are fitted to a population I invented, which is circular, and the recommendation says so where it is made rather than in a footnote.

The 2026 EDD change removes the only control catching a whole population.

SI 2026/621, made 9 June 2026, narrows mandatory jurisdiction-based enhanced due diligence to the three FATF *Call for Action* countries, rather than those plus the 22 under *Increased Monitoring*.

Had jurisdiction-based EDD before 2026	16
Have it under the 2026 rule	5
Lose it	11
Of those, escalated for any other reason	0

The model still scores them. It just cannot move them across a band. The score sees the risk; the rating does not.

The model had the control. The file did not have the input.

Santander opened a business account on a form describing the trade as a “Translation service”, expecting £5,000 a month. Within six months it was receiving millions.

Scored from the application form	1.93 → Low
With the business verified as an MSB	2.18 → High

The Final Notice records that the customer operated a money service business and that this was not identified at onboarding. Verified, the escalator fires on day one. The control was in the model the whole time.

Five documents of this project argue about weights, boundaries and aggregation rules. None of that argument survives contact with a file where the trade was never verified. A risk model is only ever as good as the due diligence feeding it, and nothing in mine measures that.

Synthetic data, a fictional bank, and a self-directed learning project alongside a BA in Accounting and Finance. Not professional advice, and not independently validated by a person. Full method, workbooks and case files on GitHub — link in the comments.